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Modal birefringence-free lithium niobate waveguides

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We investigate polarization-insensitive waveguide designs afforded by the interplay of material and waveguide birefringence in $LiNbO_3$-on-insulator photonic wires. Fundamental mode birefringence-free operation in the 0.8–1.8 μm spectral range is predicted for a suitable choice of waveguide widths in the 375–600 nm range. Optimized buried waveguide designs yield broadband (1350–1625 nm) index matching between TE_{00} and TM_{00} modes. Furthermore, simultaneous phase- and group-velocity matching at infrared wavelengths appears feasible for pulse durations as short as 100 fs.

OCIS codes: (230.7370) Waveguides; (130.3120) Integrated optics devices; (130.3730) Lithium niobate; (260.1440) Birefringence.

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Lithium niobate (LN) is an attractive material for photonic applications due to its electro-optic and nonlinear optical properties [1] and mature waveguide and quasi phase-matching technologies [2,3]. The recent emergence of thin-film LN on insulator (LNOI) platforms has revived the interest in LN, holding promise for a new generation of integrated optical switches and modulators, fully harnessing the most appealing features of this material, in ever more compact and efficient device formats.

The smaller footprints and enhanced efficiencies afforded by high-confinement integrated photonic platforms, such as LNOI or silicon on insulator (SOI), come nevertheless at the price of strong waveguide-induced birefringence. The latter induces polarization-dependent phase shifts and time delays, which can negatively affect efficiencies, extinction ratios, and bandwidths of switching and modulation devices. Such issues are particularly critical in devices interfaced with standard telecom fibers (without strict control of the input polarization state) as, e.g., LN ultrahigh-speed wavelength converters [4]. Removing the polarization dependence of sub-micrometric photonic wires, while granting single-mode operation, presents one of the most difficult challenges in the design of planar waveguide components, with a considerable portion of silicon photonics research being specifically devoted to this issue [5].

A number of solutions have been proposed for SOI photonic wires, including birefringence compensation via stress engineering in waveguide cladding layers or by introducing additional form-birefringence via advanced patterning techniques [5,6]. Here we address this issue in the case of LNOI photonic wires. At difference to SOI, in the LNOI platform additional engineering degrees of freedom are afforded by the high intrinsic birefringence of LN. With vectorial modeling of the LNOI photonic wires, we show that suitable choices of waveguide dimensions and crystal orientation enable fundamental mode birefringence-free operation in a broad spectral range, extending from 0.8 to 1.8 μm. Moreover, we identify polarization-insensitive designs affording simultaneously phase- and group-velocity matching for TE_{00} and TM_{00} guided modes, a unique feature not envisageable in other platforms such as SOI [6].

Figure 1 provides a cross-sectional view of the structures considered in this Letter, amenable to fabrication by advanced nanostructuring of commercial LNOI wafers [7,8]. The waveguide core is made of congruent LN and has a rectangular cross section of height h and width w. The waveguide lower cladding consists of a ~2 μm thick silica layer, which may be grown on a silicon [9] or LN [10] carrier wafer. The latter is not shown in the figure, since the high-confinement waveguides considered here are engineered so as to avoid mode leakage into the substrate beyond the silica layer. Figures 1(a) and 1(b) illustrate the two cases analyzed in what follows, corresponding to the ridge and buried LNOI waveguides, with upper claddings made of air and silica, respectively. In the latter case, we assumed conformal coating by a 2 μm thick SiO_2 layer.

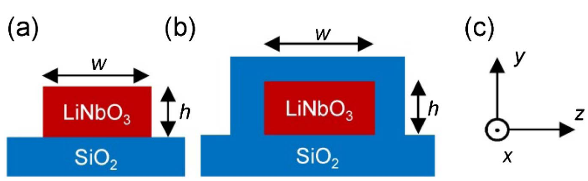

Fig. 1. Cross-sectional views of (a) ridge and (b) buried photonic wire geometries on LNOI, where h and w denote the height and width, respectively, of the LN waveguide core. (c) Orientation of the principal dielectric axes of LN with respect to the waveguide transverse plane.

Figure 1(c) shows the chosen orientation for the LN principal dielectric axes with respect to the photonic wire. The geometry corresponds to etched waveguides in Y-cut LNOI films, with transverse field confinement in the Y–Z plane and propagation along the X axis of LN, just as in Refs. [7,8]. Identical results hold true when the X and Y axes are swapped.

The basic principle employed in the design of modal birefringence-free waveguides in LNOI is first illustrated with an intuitive argument, which can be developed for buried LNOI waveguides, i.e., Fig. 1(b). Based on symmetry considerations, the geometrical condition $w = h$ (square waveguide) can then be identified as the one that grants degeneracy of the effective indices for (quasi-) TE and (quasi-) TM modes, $N_{\mathrm{TE}} = N_{\mathrm{TM}}$, when the material (LN) birefringence is neglected. When $w > h$, the waveguide birefringence keeps the effective index of the TE mode above the TM one, i.e., $N_{\mathrm{TE}} > N_{\mathrm{TM}}$ [11]. Therefore, the natural choice to counteract this effect by means of the LN intrinsic birefringence is to attribute the TE and TM modes to the extraordinary (Z) and ordinary (X, Y) polarizations, respectively, given that LN is a negative uniaxial material ($N_e < N_o$) [1,12]. This intuitive insight is confirmed by systematic vectorial simulations of the modal structure of both ridge and buried LNOI waveguides, which we performed for wavelengths in the visible and infrared, up to $\lambda \sim 2$ μm.

The guidance properties of the LNOI photonic wires were simulated with a finite-elements-based vectorial mode-solver (COMSOL 5.2), using Sellmeier equations from the literature to model the wavelength- and polarization-dependent refractive indices of congruent LN [13], as well as SiO_2 [14]. The polarization response of the LNOI photonic wires was studied for the configurations of Figs. 1(a) and 1(b) for different wavelengths (λ) and waveguide dimensions (h, w). In what follows, for the sake of simplicity, we shall present the results for a fixed waveguide height ($h = 300$ nm), which ensures single-mode guidance in the vertical direction over the targeted spectral range ($\lambda = 0.6$–1.8 μm).

The evolution of the guided-mode effective indices (N_{eff}) with the waveguide width for a fixed wavelength is illustrated in the plots of Figs. 2(a) and 2(b), with reference to the ridge and buried waveguides, respectively. The curves show the computed indices for the first six guided modes at $\lambda = 1064$ nm. For large waveguide widths ($w > 3$ μm), the effective index diagrams follow the trend of traditional large-mode-area LN waveguides [11], where the TE indices exceed the TM ones, tending to the asymptotic limit of a slab waveguide. On the other hand, for decreasing widths of the LN waveguide core, the waveguide dispersion becomes progressively stronger, altering the balance between the mode indices of the two polarizations. This eventually results in the appearance of crossing points between the effective index curves, highlighted by the circles in Fig. 2, corresponding to critical waveguide widths for which TE- and TM-mode indices are equalized. As shown by Fig. 2, the first effective index equalization is achieved with a high-order TE mode, i.e., TE_{30}, for a waveguide width $w = 2.89$ μm (3.69 μm) in the ridge (buried) configuration. Polarization-insensitive operation for the two lowest-order guided modes, $N_{\mathrm{eff}}^{\mathrm{TM00}} = N_{\mathrm{eff}}^{\mathrm{TE00}}$, occurs instead at $w = 416$ nm (398 nm). The intrinsic birefringence of LN, however, implies that such an operation cannot be reached in a square waveguide ($w = h$), as shown by the insets of Figs. 2(a) and 2(b).

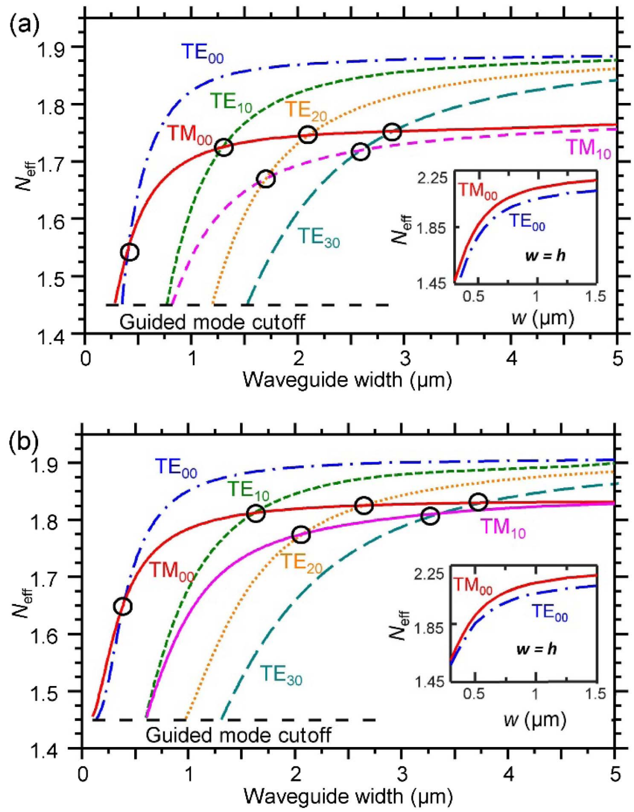

Fig. 2. Effective indices (N_{eff}) of the first six guided modes calculated as a function of the waveguide width (w) for a waveguide height $h = 300$ nm and a wavelength $\lambda = 1064$ nm, in case of (a) ridge and (b) buried waveguides (Fig. 1). The circles highlight the points where two orthogonally polarized modes experience the same effective index. Insets: effective indices (N_{eff}) of the fundamental modes of a square waveguide ($w = h$) as a function of its side length.

The comparison between the simulations for the ridge and buried waveguides indicates an overall better confinement attained in the latter, where modal effective indices are overall higher. One implication is that at the zero-birefringence points for their fundamental modes (TE_{00}-TM_{00} index matching), the SiO_2-cladded waveguides are further away from cutoff than their ridge counterparts. Furthermore, the buried waveguides support good guidance at longer wavelengths than ridge ones, exhibiting an operational range ultimately limited only by the absorption edge of silica ($\lambda \sim 2$ μm). The mode profiles of the buried waveguides are also much more symmetric, as illustrated by the inset of Fig. 3 (showing the modulus of the electric field components polarized along Z and Y, for TE_{00} and TM_{00} modes, respectively). This feature maximizes the modal spatial overlaps as well as the coupling efficiencies with optical fibers. In light of all such features, the buried waveguide configuration is the most appealing for practical devices; hence, the rest of the discussions shall focus on this geometry.

Figure 3 illustrates the evolution of the TE_{00} and TM_{00} mode effective indices as a function of the wavelength for a 398 nm wide buried waveguide [TE_{00}-TM_{00} crossing point of Fig. 2(b)]. The effective index curves closely follow each other in a wide wavelength range, even beyond $\lambda = 1064$ nm, reaching out deeper in the infrared. This is due to the existence

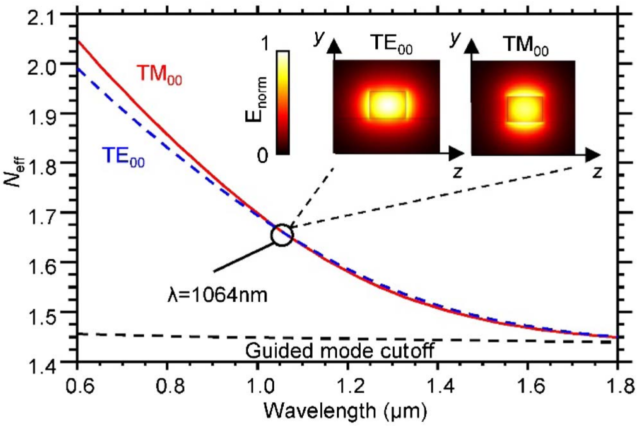

Fig. 3. Evolution of the effective indices of the TE_{00} and TM_{00} guided modes in a 300 nm high and 398 nm wide buried LNOI waveguide as a function of the wavelength. Inset: normalized electric field distributions, $E_{\mathrm{norm}}(z, y) = |E/E_{\max}|$, for TE_{00} and TM_{00} modes (polarized along z and y, respectively) at the zero-birefringence point.

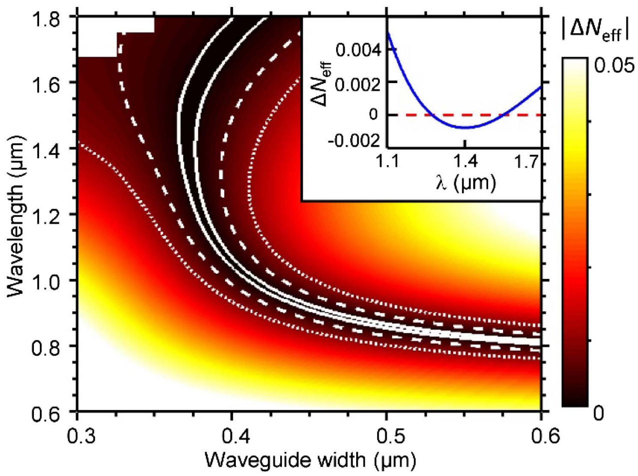

Fig. 4. 2D plot of the fundamental mode birefringence $|\Delta N_{\mathrm{eff}}| = |N_{\mathrm{eff}}^{\mathrm{TM00}} - N_{\mathrm{eff}}^{\mathrm{TE00}}|$ as a function of the waveguide width (w) and wavelength (λ), for a buried LNOI waveguide with $h = 300$ nm. The solid, dashed, and dotted contour lines represent the loci of points where $|\Delta N_{\mathrm{eff}}| = 0.001$, 0.005, and 0.01, respectively. The white area on the top left corner corresponds to the guided mode cutoff. Inset: ΔN_{eff} versus wavelength for $w = 376$ nm.

of a second spectral point where the indices of the TE_{00} and TM_{00} modes are exactly matched (occurring at 1813 nm in this case). As a consequence, modal birefringence-free operation can be expected to occur in a broad spectral range. This aspect was further investigated with extensive numerical investigations.

The contour map of Fig. 4 shows the dependence of the TE_{00}-TM_{00} mode birefringence on the waveguide width and the optical wavelength, for a buried LNOI waveguide with $h = 300$ nm. As a figure of merit for the birefringence, we take here the modulus of the effective index difference between the two modes $|\Delta N_{\mathrm{eff}}(\lambda, w)| = |N_{\mathrm{eff}}^{\mathrm{TM00}} - N_{\mathrm{eff}}^{\mathrm{TE00}}|$. The map clearly indicates the possibility to achieve birefringence-free operation across the full 0.8–1.8 μm spectral range by a suitable choice of the waveguide width in the 375–600 nm range.

A particularly interesting feature highlighted by Fig. 4 is the presence of a turning point in the modal zero-birefringence curve in the $\{\lambda - w\}$ plane, located at a waveguide width $w \sim 376$ nm. As a consequence, broadband $TE_{00} - TM_{00}$ index matching with a wide wavelength coverage, from the E- to the SCL-telecom bands (1360–1625 nm), can be achieved by design. This feature is further illustrated through the inset of Fig. 4, where the index difference ΔN_{eff} is plotted as a function of the wavelength for this particular waveguide width. On the other hand, for wider waveguides ($w > 0.7$ μm) the zero-birefringence wavelength (753 nm) is not tunable with waveguide dimensions and features a much narrower spectral acceptance bandwidth (~9 nm).

In order to assess the practical viability of the birefringence-free LNOI designs, it is also important to estimate their sensitivity to variations in the waveguide width that may be induced by fabrication imperfections. To this aim, on the color plot of Fig. 4, we superimposed contour lines corresponding to three specific values of $|\Delta N_{\mathrm{eff}}|$, namely 0.01, 0.005, and 0.001, plotted as solid, dashed, and dotted lines, respectively. To provide a quantitative estimate, let us consider the working point of Fig. 3, i.e., $\lambda = 1064$ nm and $w = 398$ nm. Maximum allowed birefringence values ($|\Delta N_{\mathrm{eff}}|$) amounting to 0.01, 0.005, and 0.001, yield tolerances against width variations (Δw) of 64, 32, and 6 nm, respectively. Larger tolerances are expected at telecom wavelengths. For instance, for the operation point at $\lambda = 1550$ nm ($w = 376$ nm), the width tolerances corresponding to the three above-listed values of $|\Delta N_{\mathrm{eff}}|$ become $\Delta w = 160$, 36, and 14 nm, respectively. Similar values are found for the waveguide height tolerances, e.g., $\Delta h = 14$ nm for $|\Delta N_{\mathrm{eff}}| = 0.001$ and $|\Delta N_{\mathrm{eff}}| < 0.005$ for $h < 325$ nm.

For a number of applications, such as all-optical switching, wavelength conversion, or entangled photon generation, it may be also important to achieve polarization-insensitive operation in the pulsed regime. A feature of merit for the performance in this case is provided by the temporal walk-off acquired through propagation over the device length (L) by the pulse components travelling with modes of orthogonal polarization. Such a temporal walk-off is directly proportional to the group-index difference (ΔN_g) at the pulse center wavelength λ. Therefore, we chose the quantity $\Delta N_g = N_g^{\mathrm{TM00}} – N_g^{\mathrm{TE00}}$ as a figure of merit for the TE_{00}-TM_{00} group-index birefringence, where the modal group index N_g is given by $N_g = N_{\mathrm{eff}} – \lambda \partial N_{\mathrm{eff}} / \partial \lambda$ and, hence, can be determined from the computed wavelength-dependent modal effective index (N_{eff}). The knowledge of ΔN_g enables the direct evaluation of the minimum pulse duration for walk-off-free operation in a device of length L, given by $\Delta \tau = L \Delta N_g / c$, where c is the speed of light.

In Fig. 5, we show the computed values of ΔN_g for a 300 nm-high buried LNOI waveguide, as a function of the wavelength and waveguide width. The solid contour lines superimposed on the 2D plot delimit the region where $|\Delta N_g| < 0.001$. From Fig. 5, it is apparent that TE_{00} to TM_{00} group-velocity matching can be achieved at essentially any wavelength from 1.176 up to ~1.7 μm, through a suitable choice of the waveguide width in the 330–600 nm range. The largest spectral acceptance bandwidth amounts to $\Delta \lambda = 34$ nm and is obtained around $\lambda = 1683$ nm for $w = 335$ nm

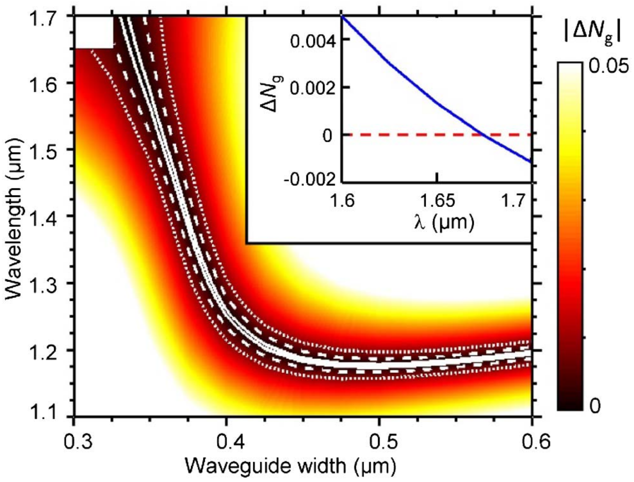

Fig. 5. 2D plot of the computed TM_{00}-TE_{00} group-index difference $|\Delta N_g| = |N_g^{TM00} - N_g^{TE00}|$ as a function of waveguide width (w) and wavelength (λ), for a buried LNOI waveguide with $h = 300$ nm. Solid, dashed, and dotted contour lines represent $|\Delta N_g| = 0.001$, 0.005, and 0.01, respectively. The white area on the top left corner corresponds to the guided mode cutoff. Inset: ΔN_g versus wavelength for $w = 335$ nm.

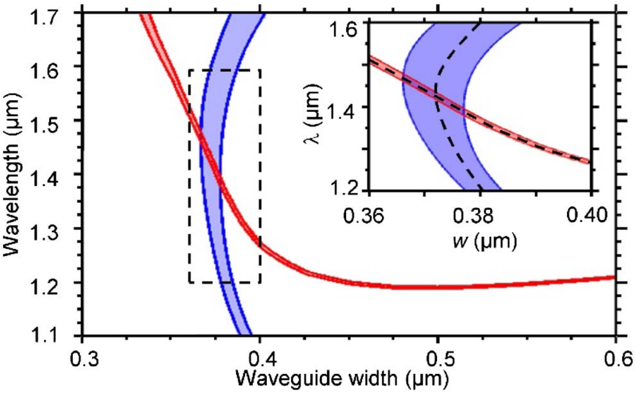

Fig. 6. 2D diagram highlighting the regions for TM_{00}-TE_{00} group-index and effective index matching in buried LNOI photonic wires with $h = 300$ nm. The red and blue regions represent the loci of points where $|\Delta N_g|$ and $|\Delta N_{\mathrm{eff}}|$, respectively, are lower than 10^{-3}. Inset: zoom-in of the region around $\Delta N_g = \Delta N_{\mathrm{eff}} = 0$.

(inset of Fig. 5). This would correspond to walk-off-free operation with a pulsewidth $\Delta\tau \sim 100$ fs for a propagation length $L \sim 3$ cm. The tolerances versus waveguide fabrication errors are somewhat tighter than in the case of matching the effective indices. Nevertheless, they can still fit within realistic constraints in mm-long devices, e.g., $\Delta w = 33$ nm for $\Delta\tau = 100$ fs at the optimum point for broadband operation ($\lambda = 1683$ nm), if $L = 3$ mm ($\Delta N_g = 10^{-3}$).

Finally, Figs. 4 and 5 allow to recognize the existence of a point where zero-birefringence and group-velocity matching can be simultaneously attained, occurring for $\lambda = 1429$ nm and $w = 372$ nm (Fig. 6). In this work point, birefringence- and walk-off-free operation is expected for pulse durations down to $\Delta\tau \sim 30$ fs, in a 1 cm-long device. This peculiar feature has not been highlighted so far in any other equivalent photonic platform.

In conclusion, we theoretically analyzed configurations to implement mode birefringence-free high-confinement waveguides on the thin-film LNOI platform, taking advantage of the high LN birefringence to partially alleviate geometrical constraints. We highlighted the possibility to attain fundamental mode birefringence-free operation for any wavelength in the 0.8–1.8 μm spectral range by choosing waveguide widths in the 0.35–0.6 μm range. Broadband polarization-insensitive operation extending beyond the telecom band (1360–1625 nm) is also predicted with LNOI photonic wires of a 376 nm × 300 nm cross section. Similar design criteria were applied to investigate group-velocity matching among fundamental modes of orthogonal polarization. A remarkable feature emerging from the analysis is the design of LNOI waveguides where modal birefringence-free and group-velocity matching can simultaneously be achieved.

Anticipating potential challenges in the control of waveguide dimensions and non-vertical sidewalls of the fabricated devices, the accepted tolerances in width variations were also studied. The proposed designs appear fully feasible in light of the most recent developments of integrated optics technology on LNOI [7,8,12], and can enable advanced polarization-control functionalities in electro-optic and all-optical switching devices, micro-resonators, and nonlinear frequency converters now coming of age on the LNOI integrated photonics platform.

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